

Language Death and Endangerment: An Interactive Visualization of the Status of Dead, Dying, and Endangered Languages Worldwide

CSC-475 Project Proposal

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Abstract

Despite the attention given to language death by linguists in recent decades, the general population is largely unaware of language death and its implications. Furthermore, even experts must navigate paywalls to access conventionally creditable works, such as *Ethnologue* [1]. To address issues of accessibility and awareness, a dataset compiling publicly available information is provided, coupled with an analysis of the availability of that information. This data is then used to provide a website visualizing this information, providing a general audience the opportunity to explore the issue of language death in a palatable format.

1 Introduction

Language death is not a novel topic in the field of linguistics. Rather, the phrase *language death* was first used in 1972, lending the field the opportunity to discuss, research, and preserve language status for the past 50 years. [2] Yet, despite humanity's current disposition to lose over 6,000 languages within the century - and with them cultures, knowledge, identities, and social customs - the reality of language death, and the implications therein, are seldom known outside of the field.

The grim outlook of world languages is effectively conveyed by the following estimates:

1. The Language Conservancy (TLC) estimates that "more than 7,000 languages resound across our planet every moment, but about 2,900 or 41% are endangered. At current rates, about 90% of all languages will become extinct in the next 100 years." [3]
2. The United Nations Educational, Scientific and Cultural Organization (UNESCO), estimates an even faster rate of language death, positing that "around 3,000 languages could disappear before the end of the century, at a rate of one every two weeks." [4]

Languages encapsulate much more than semantics: they carry with them the history, experiences, and ways of life of their speakers. This ranges from indigenous medicinal practices - many of which use flora native to the land, often untranslated and unrecorded in any other language - to nuanced phrases developed from the experiences of generations.

Despite the gravity of such losses, the most comprehensive literature exists behind paywalls. Conventionally credible sources such as *Ethnologue* and *Atlas of the World's Languages* require even academics to pay lofty subscription and purchase fees, occasionally copyright licenses. [1] [5] Furthermore, resources such as UNESCO's The World Atlas of Languages (UNESCO WAL), which advertise downloadable data to account holders - obtainable through signing up for an account - is

poorly monitored.¹ This leaves the general population with access to *one* comprehensive resource: *Glottolog*. [6]

Glottolog, although publicly accessible, is subject to three criticisms:

1. *Glottolog* is not considered the conventional source for information about languages, and thus is not perceived within academia as credible when compared with the standard: *Ethnologue*.
2. *Glottolog* data is not available for download in its entirety. Although this is common, and *Glottolog* creators are not obligated to release their hard-earned data, this leaves people with no publicly available resource for viewing comprehensive language data.
3. *Glottolog* is not easily navigable for all users. Although this is surely a qualitative assessment, it is worth considering that *Glottolog* windows are nested, and require users to navigate through multiple steps to view statistics on a single language. For the general population, this organization may not be maximally effective.

These assessments are not intended to be critical; rather, they are meant to assess the usability of *Glottolog* in raising awareness of language death when presented to the average user. Although comprehensive, the navigability of *Glottolog* is a potentially ineffective presentation of language data.

To address the issue of accessibility described above, I propose the creation of a website which has the primary function of:

1. The exclusive visualization of *publicly available* information about languages.
2. A visualization which emphasizes the extent to which languages are endangered.
3. A visualization which emphasizes the *lack* of publicly available data on language status.

This was accomplished using minimal information from *Glottolog*, and the majority of information from *Wikipedia* language pages.²

2 Related Work

The modern field of Linguistics emerged from that of Philology in 1837, alongside Historical Linguistics. The earlier term, philology, describes "the branch of knowledge that deals with the structure, historical development, and relationships of languages or language families." [7][8] This discipline, although fundamentally equivalent to the modern field of linguistics, lacked any distinction between the comprehensive discipline and any sub-disciplines therein. In 1837, concurrent with the emergence of the term linguistics, the emergence of sub-disciplines became apparent; particularly that of Historical Linguistics, which is expressly concerned with the history and development of languages. [9] During the early 19th century, scholars endeavored to establish languages families and reconstruct unattested proto-languages. A pointed example of this trend may be seen in the surge of research surrounding the Indo-European language phylum at this time.

The field of Linguistics did not see the emergence of any further major sub-disciplines until the early 20th century, when the field of Sociolinguistics was first postulated. Sociolinguistics investigates the social environment through which languages change, and thus reconvened scholars around the question of what prompts language change, as opposed to the previously exclusive inquiry of *how* languages have changed. [10] This introduced a number of novel perspectives to the field of linguistics, and has brought with it an abundance of observations, including the investigation of the phenomenon now known as language death, which was first proposed to the field by Dressler in 1972. [11]³

¹The author and their university both applied for an account, both of which have been pending approval from UNESCO WAL for approximately four months.

²For more information regarding what data was collected from what sources, see the description of data acquisition in the Methods section.

³While many sub-disciplines have emerged in the field since then, they will not be reviewed herein, as they do not pertain to the contents of this paper.

Language death describes the "process or phenomenon whereby a language, usually that of a cultural minority, disappears or falls into disuse" [12] [11] In 1972, Austrina Linguist Wolfgang U. Dressler proposed that

Language shift⁴ and language death are the endpoints of a sociolinguistic development affecting minority languages. In a situation of competition between an underprivileged minority language⁵ and at least one dominant language, sociolinguistic norms of language choice may change in such a way that finally a total shift away from the minority language may take place. ... Accordingly, the fundamental task of a linguist working on language decay and language death is to identify, to describe and to explain linguistic phenomena of language decay and language death. [16]

In response, sociolinguists have begun analyzing languages, and the circumstances surrounding their perception and use - often discovering a variety of social factors leading to language attrition.⁶

In a continuation of his previous work, Dressler has published a holistic review of the various causes of language decay and language death he observed during his career in his *Independent, Dependent and Interdependent Variables in Language Decay and Language Death*. [18] Dressler articulates the current perception of the field aptly:

Within the history of linguistics, systematic investigations in this area [language death] started only in the 1970s, but have aroused new interest in this century due to a new focus on language revitalisation and language revival as well as due to greater awareness of how many languages are in danger of vanishing. Krauss reckoned with the imminent disappearance of between 20% and 50% of all contemporary languages and with a final survival of only 10%. Moseley's *UNESCO Atlas of the World's Languages in Danger* considers 43% of all contemporary languages to be endangered, of which, in a scale of endangerment, 574 languages are considered to be critically endangered, 529 severely endangered, 645 definitely endangered and 595 vulnerable. [19] [20]

As Dressler states in his work, there are a number of causes of language death that are *not* covered in this article, which are characterized as 'sudden death' or 'radical death' causes (i.e. "genocide and connected linguicide"⁷ [18]). Due to the constraints of space in this paper, the following summary is proposed by Dressler with his own acknowledgement of its oversimplification:

A simplified causal chain frequently starts with socio-political events of, directly, national language policies (especially in education); and indirectly with imperialism, centralisation and/or nationalism,... This leads to marginalisation, assimilation, repression of large sections of speakers of minority languages. But this may also lead to resistance against oppression, as in the case of Catalan,... Contributing factors are co-territoriality instead of spatial separation,... and the demography of relatively small numbers of speakers, their mean age and percentage of fluent young speakers.

A second level, often dependent on the socio-political one, is the socio-economic level. Its main elements are economic pressure via economic subordination, urbanisation, industrialisation, and greater geographical and social mobility.

A third level, dependent on both or one of the preceding levels, is socio-cultural.

⁴Language Shift is defined as "the process whereby an individual or speech community moves from habitually using one language to using another, often culturally dominant, language; an instance of this." [13] [14]

⁵Minority Language is defined as "a language spoken by a minority group, if different from that of the majority." [15]

⁶Language Attrition is defined as the "loss of ability in a language, or gradual decline of a language, esp. in a bilingual or multilingual community." [17]

⁷"There are many different types of language decay and language death that are not covered in this contribution, such as genocide and connected linguicide, i.e. by killing all or nearly all native speakers of a language, and other types of 'sudden death' or 'radical death', which are not preceded by a period of language decay, language attrition of immigrant heritage languages or of second language competence." [18]

Here the keywords are acculturation, racism, folklorisation, loss of prestige, perceived small linguistic distance between recessive and dominant language, and limited use in the media, where now the new media give a new chance to recessive languages marginalised in the print and audiovisual media.

A fourth level, depending on the preceding ones, is the sometimes forgotten mediating socio-psychological level, with the dimensions of internal and external attitudes toward dominant versus recessive language, degree of language loyalty, and lack of identification with the recessive language.

Linguistic factors, belonging to a fifth level, are first of all dependent on the previous levels,... In particular, a decaying language may be difficult to identify with and to be loyal to. Thus, older generations may despise young generations' usage. The decaying language may become more and more dysfunctional within the domains of communicative competence, not only by the reduction of functional domains where the recessive language is used, but also by less efficient use of it in the remaining domains.

All this leads ultimately to the final so-called 'tip' which dooms the decaying language, i.e. to the unwillingness of parents to successfully transmit the decaying language to their children and to the children's unwillingness to acquire it or to adolescent discontinuation of regular use. This leads to a break in the inter-generational transmission of the recessive language. [18]

Following Dressler's pioneering works, scholars have sought to analyze language decay through a variety of lenses. Some scholars, such as Kornai, stress the severity of language loss during a digital age, position that "of the 7,000 languages still alive, perhaps 2,500 will survive, in the classical sense, for another century. With only 250 digital survivors, all others must inevitably drift towards digital heritage status (Nynorsk) or digital extinction (Mandinka). This makes language preservation projects such as <http://www.endangeredlanguages.com> even more important." [21] Others take a more optimistic tone, proposing that the best way to study language loss is "is to study language vitality and to unveil the key elements that support it." [22] Simultaneously, scholars like Samar and Bhatia expand on individual factors of language loss, as proposed by Dressler, and present a "tentative hypothesis that the surface structural compatibility of the two languages in contact may lead to drastic changes and the possible death of the less dominant and less prestigious language." [23]

However, it is scholars like Chebet-Choge, who attempts to capture the *significance* of language loss, that speak most to the general population. In their study of Nandi anthroponyms, Chebet-Choge posits that "although traditionally the Nandi naming system is elaborate and spans a person's lifetime, present day naming practices show that various types of anthroponyms have low usage and others are no longer used. Since the Nandi naming system is intertwined with the Nandi way of life, anthroponyms act as catalogues of past and present Nandi histories. Therefore, when certain anthroponyms die, certain forms of cultural knowledge and history embedded in them are also lost." [24] The loss of linguistic and cultural knowledge poses a serious threat to minority groups, especially those that stand to lose histories passed down through oral tradition. This reality brings about the question of awareness, and that of disseminating information regarding language loss.

3 Data

To emphasize what information is publicly available for world languages, the following information was obtained from *Wikipedia* pages (where discover-able) for a complete list of world languages as provided by ISO-639-3 Codes:

1. The name of each language in English (as labeled by ISO Codes),
2. The ISO-6393 Code,
3. The glottocode,
4. The degree of endangerment as listed by *Glottolog*'s Agglomerated Endagerment status,
5. The degree of endangerment as listed by UNESCO WAL,
6. The level of the language as listed by *Glottolog*,

7. The macroarea of the language as listed by *Glottolog*,
8. The type of language as listed by *Glottolog*,
9. The scope of the language as listed by *Glottolog*,
10. The number of speakers as listed by *UNESCO WAL* where applicable, and *Wikipedia* elsewhere,
11. The dialects as listed by *Wikipedia*,
12. The regions where a language is spoken as listed by *Wikipedia*,
13. The nations where a language is an official language as listed by *Wikipedia*,
14. The nations where a language is a recognized minority language as listed by *Wikipedia*,
15. The latitude value for a language as listed by *Glottolog*,
16. The longitude value for a language as listed by *Glottolog*, and
17. The url used to scrape information from *Wikipedia* regarding that language.

ISO-639-3 Codes were downloaded from the woorm/iso-639-3 github repository. [25] Values obtained from *Glottolog* were pulled from their downloadable files found at <https://glottolog.org/meta/downloads>. [6] Values obtained from *Wikipedia* were scraped from the infobox table on a given page. ⁸ Values obtained from UNESCO were scraped from the data provided on the page found at <https://www.theguardian.com/news/datablog/2011/apr/15/language-extinct-endangered>. [26]

While url's for each language were not sought out manually for all columns, Figure 1 displays the percentage of rows for each column value wherein data was not obtainable from a wikipedia page, which often corresponds to whether or not a url was or was not provided. This visualization provides a pointed display of the unavailability of language data, as *Wikipedia* often provides a thorough (although sometimes out of date) review of publicly available information.

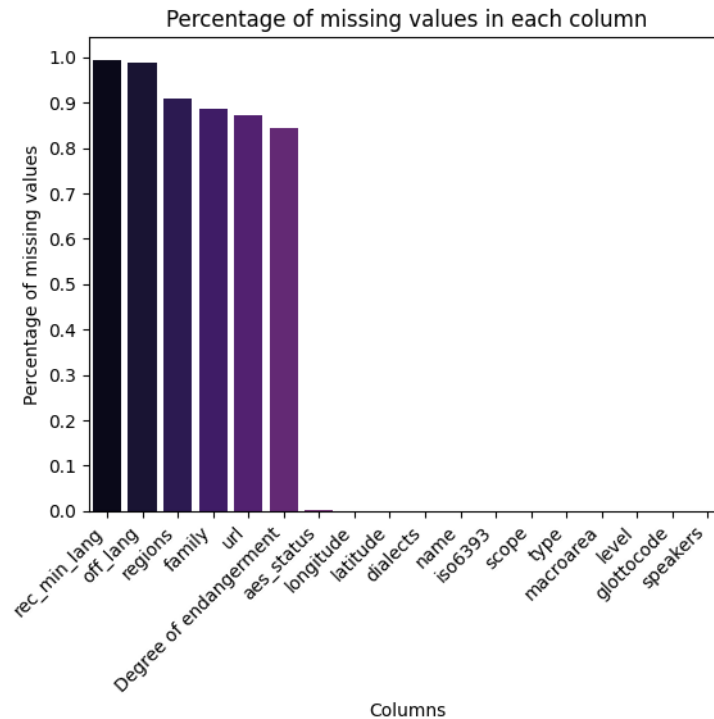


Figure 1: Visualization of unpopulated rows by columns as a percentage.

⁸More information about infobox tables can be found at <https://en.wikipedia.org/wiki/Help:Infobox>.

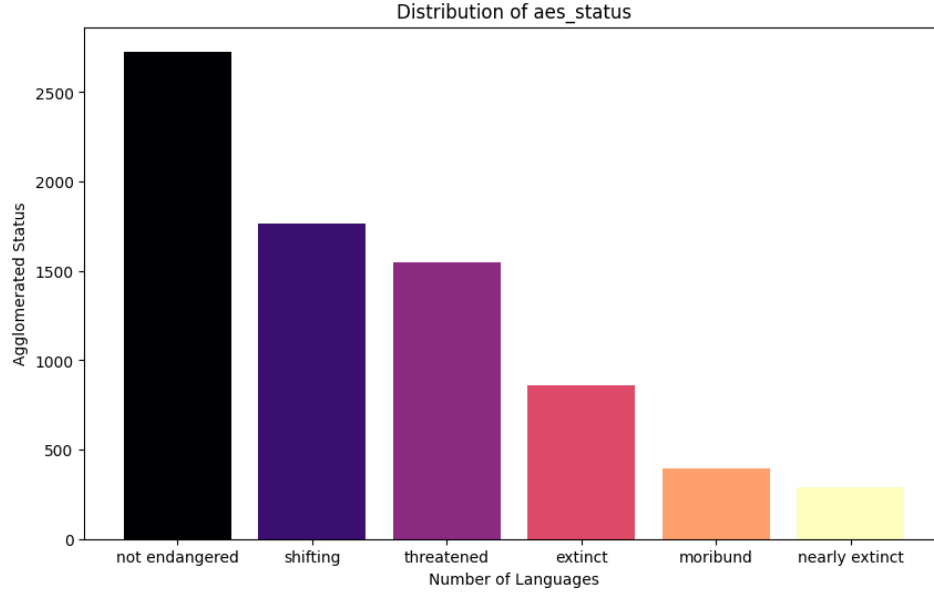


Figure 2: Distribution of language across Agglomerated Language Status column.

4 Methods

ISO-639-3 Code Data ISO-639-3 Codes are standard codes published by the International Organization for Standardization. Codes are designated to a variety of entities (i.e. nations, regions, etc), including languages. The ISO-639-3 Codes used herein were acquired via the download of the `iso6393.js` file found at the `woorm/iso-639-3` github repository. [25] This file was then converted to a `.json` file manually through the use of regular expressions, such that it could be read using the Pandas library `.read_json()` function. No further changes were necessary.

Glottolog Data *Glottolog* data was obtained via downloading the `glottolog_languoid.csv.zip` [569.5KB] file on the <https://glottolog.org/meta/downloads> page. No further changes were necessary.

The Agglomerated Endangerment Status of each language, as listed by *Glottolog* was obtained using a web scraper developed for the purpose of this paper which generated the url for every language which had an ISO-639-3 Code (which is populated for the entire dataset), and acquired the status from the Endangerment textbox on that page. The *Wikipedia* page url for each language was also obtained using the same web scraper, acquired from the *Links* sidebar on the right side of each page.

UNESCO Data UNESCO data was acquired for the speakers and `aes_status` columns by using a web scraper developed for the purpose of this paper to collect information from the Data Summary table found at <https://www.theguardian.com/news/datablog/2011/apr/15/language-extinct-endangered>. [26]

Wikipedia Data *Wikipedia* data was acquired using a web scraper developed for the purpose of this paper, which collected all of the labels found in the infobox of a given page, and then extracted the data text for any of the following labels, if present:

1. Official language in
2. Recognised minority language in
3. ISO 639-3
4. Glottolog
5. Speakers *or* Native speakers

6. Region *or* Native region
7. Language family
8. Dialects

This was done for all urls scraped from *Wikipedia*'s List of Language Names page. However, this list does not contain urls for every language page on *Wikipedia*. Thus, a web scraper was written to generate the most likely url for a Wikipedia page describing each language in the UNESCO data found in the Data Summary table found at <https://www.theguardian.com/news/datablog/2011/apr/15/language-extinct-endangered>. [26] Any invalid links were replaced with valid links manually.

Aggregating Data After collecting all of the data listed above, all dataframes generated until this point were merged. This process followed the pipeline below:

1. ISO-639-3 Data and *Wikipedia* Data was merged. Duplicates in the Name column were dropped, and ISO-639-3 Data was kept over *Wikipedia* Data for all rows and columns where populated.
2. The merged dataframe from the previous step was merged with the UNESCO Data, keeping UNESCO Data for all rows and columns where populated.
3. The merged dataframe from the previous step was merged with the *Glottolog* glottolog_languoid.csv.zip [569.5KB] Data, and *Glottolog* Data was kept for all rows and columns where populated.
4. Any rows which did not have a Glottocode were dropped.
5. Where necessary, column values were modified such that they could be read as html code that produced the desired format.

Visualizing Data To achieve a visualization wherein attention is drawn to languages which are *more* threatened, each language was visualized on a map - located at the latitude and longitude provided in the *Glottolog* Data - using a circle pointer, which were designed as follows:

Agglomerated Language Status	Color	Size
Extinct	Black	30
Nearly Extinct	Purple	25
Moribund	Red	20
Shifting	Orange	15
Threatened	Yellow	10
Not Threatened	Green	5

Table 1: Table detailing how Agglomerated Language Status [6] was represented using circle pointers on the created website.

The creation of this website was achieved using the Google Maps API⁹, HTML¹⁰, JavaScript¹¹, and CSS¹². By increasing the size of the pointers for each language as their status worsened, and shifting the colors from green to red, viewers will be able to view languages with the worst status first. Thus, the first impression of the website is intended to provide a comprehensive impression of the number of languages which are endangered, where they are located, and the proportion of languages assigned to each status. This can be seen in Figure 3. Although this visualization is admittedly cluttered, this design has been chosen intentionally, to evoke a reaction from the user.

⁹The Google Maps API was used to create a map visualization, and to plot latitude and longitude values.

¹⁰HTML was used to write a legend - seen on the left of the page - listing the Agglomerated Language Status and the color associated with it. HTML was also used for the tables detailing relevant information about each language, which the user may view by clicking a circle pointer.

¹¹JavaScript was used to visualize the circle pointers, and to connect HTML files to the page.

¹²CSS was used to style the page.

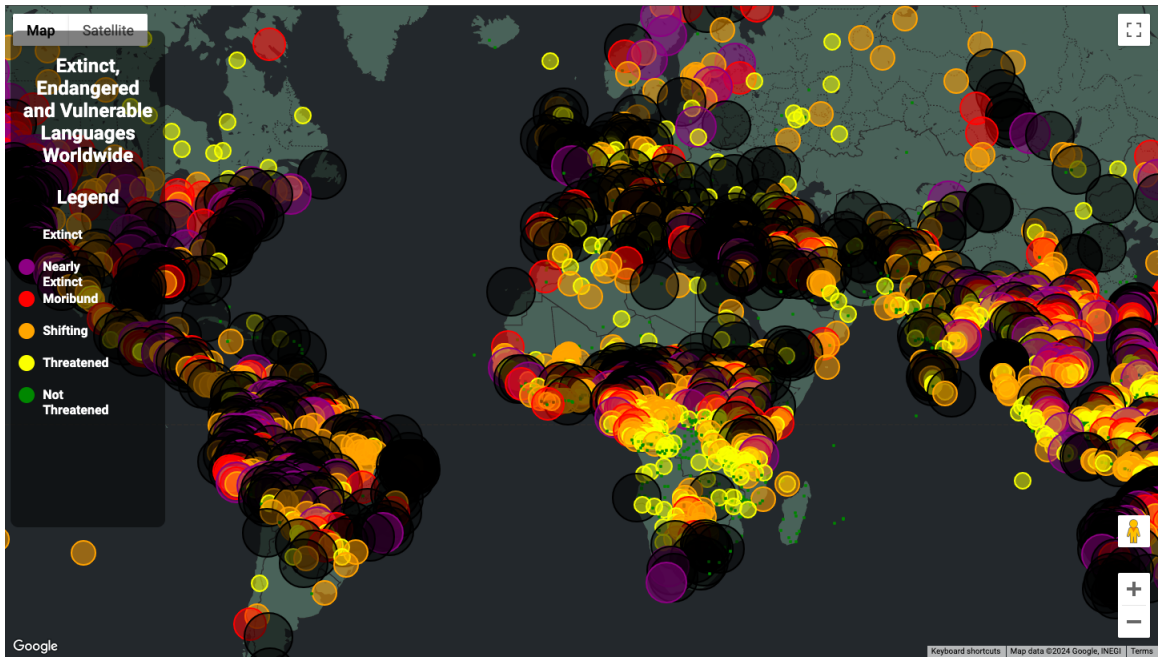


Figure 3: The default page the visualization website loads to.

To provide users with further information about a given language, each pointer displays the name of the language in English on hover. If the user clicks on a given language, information regarding the name (*Wikipedia* page hyperlink embedded), number of speakers, degree of endangerment, family, macroarea, regions, glottocode (*Glottolog* page hyperlink embedded), and ISO-639-3 code are provided (see Figure 4).

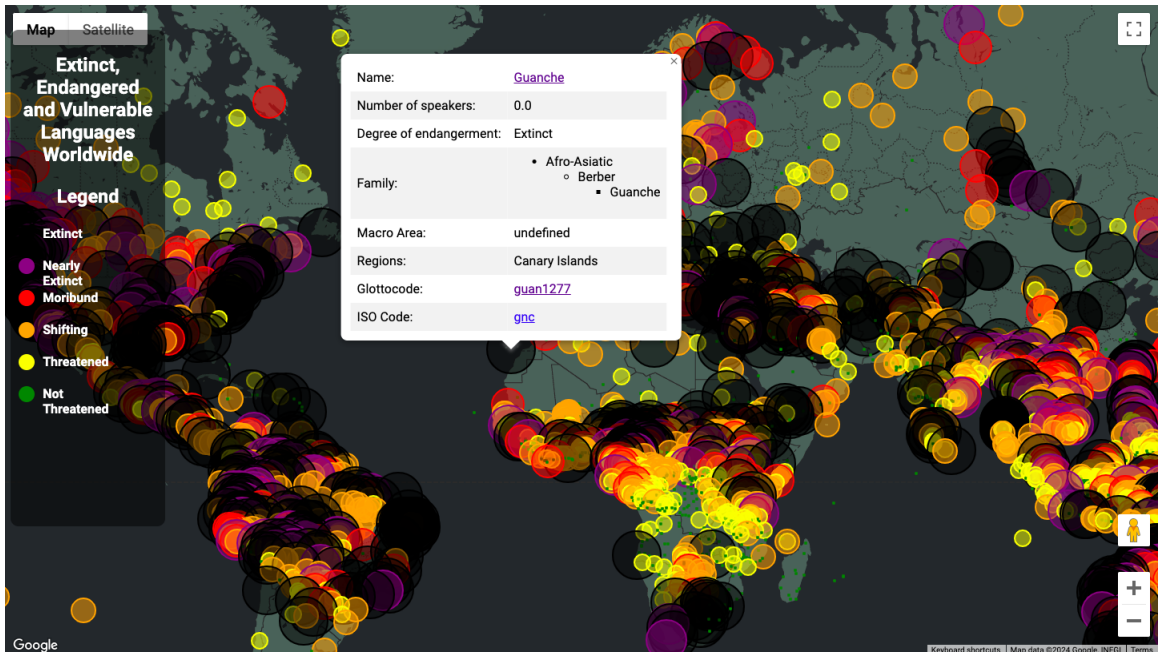


Figure 4: Sample of the table provided for each pointer on click.

5 Results

The results of this paper is the website <https://gcwarren.github.io/LanguageStatusVisualizationProject/>, which visualizes the data collected as described in the Methods section. If desired, more detailed information regarding the code used to accomplish this can be found at this github page: <https://github.com/gcwarren/LanguageStatusVisualizationProject>.

To assess the effectiveness of this website, a short survey was distributed to the author's Computer Science Seminar course, as well as the author's colleagues and acquaintances. This survey was designed to assess users' understanding - and their confidence therein - of language endangerment. Thus, students were asked the answer the questions found in https://docs.google.com/forms/d/e/1FAIpQLSeYmUz3vJR59IjwcGKlZNCyAuyXXMuZ3FXGE-RfgC88GF3JIA/viewform?usp=sf_link both *before* and *after* viewing the provided website.¹³

Of the questions asked, four were qualitative. Thus, responses made by students both before and after viewing the website are shown in the following figures: 5678. These show that, despite my website being outdated at the time of distribution, users' answers on average were correct more often *after* viewing the website. Thus, these figures show that the created website herein has achieved the desired effect of giving a general audience access to an impressionable presentation of language endangerment.

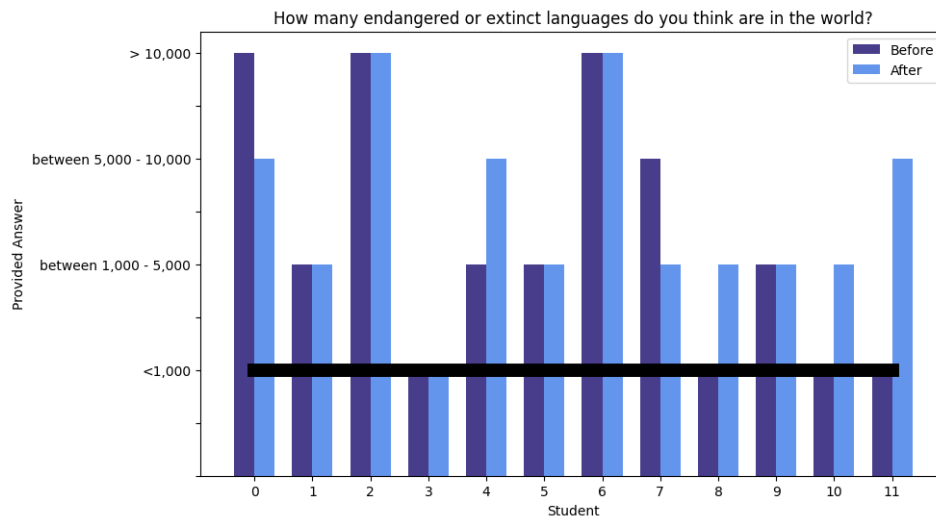


Figure 5: Student responses concerning how many endangered languages are currently in the world.

¹³It may be noted that at the time of the survey, $\approx 1,700$ languages were being visualized at that time. These languages were being visualized using UNESCO's Language Endangerment Index, and were taken from the table provided in UNESCO's Endangered Language dataset. [26] Thus, no safe (or not threatened) languages were being visualized at the time, and the website they were viewing was less detailed than what is provided herein.

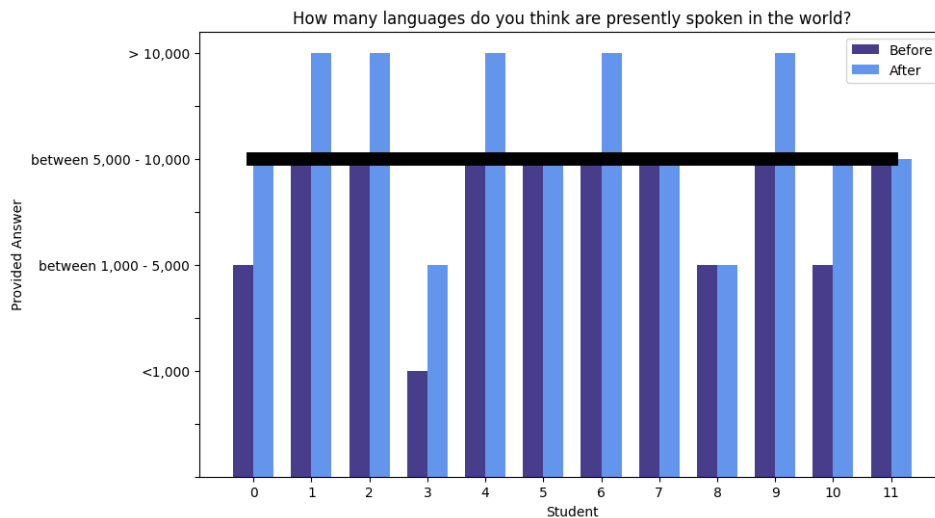


Figure 6: Student responses concerning how many languages are currently in the world.

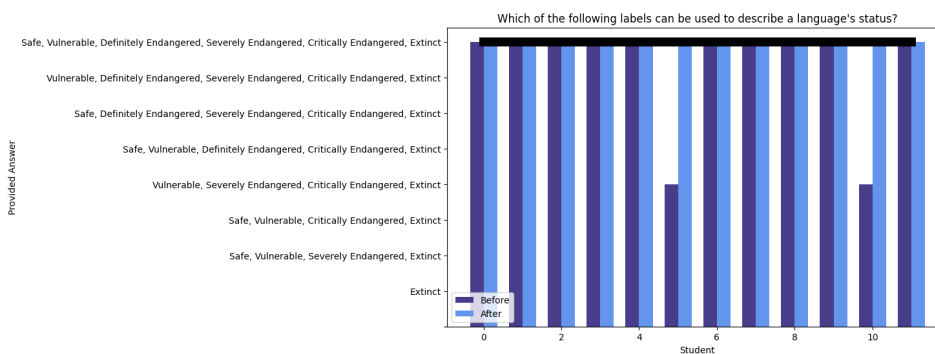


Figure 7: Student responses concerning UNESCO Language Endangerment labels.

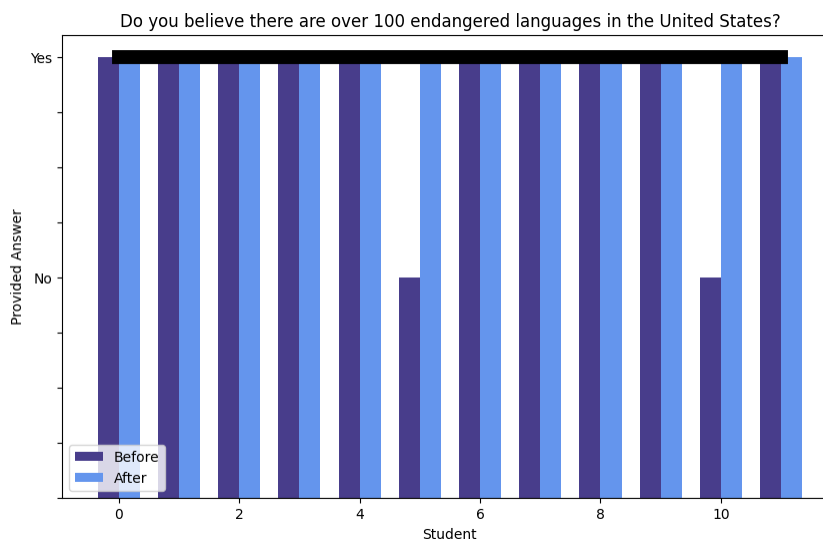


Figure 8: Student responses concerning the number of endangered languages in the United States.

6 Future Work

As displayed by this paper, information regarding language death is not always publicly accessible. That which is can sometimes be difficult to navigate, and comprehensive downloadable data often impossible to find.

Presently, humanity is faced with the loss of 90% of world languages within the next century. Yet, efforts to make information regarding this issue more accessible are low, and ultimately fall short. This paper provides a compilation of as much publicly available information as possible. Yet, this information is not guaranteed to be recent, nor is it guaranteed to be as accurate nor as detailed as the data which lies behind *Ethnologue*'s paywalls. However, the data provided herein is public, and individuals who wish to utilize it are free to.

Future uses of this data include improving the accuracy and resolution of the data herein, and implementing a downloadable feature to the website itself, such that users do not have to visit the github repository page to attain the collected data.

The optimal outcome of this paper would be the dissemination of more detailed data from organizations such as *Ethnologue*, *Glottolog*, and *UNESCO WAL*. Additionally, introducing the topic of language death - and its implications - to a general audience through social media, documentaries, and other promotional mediums would raise further awareness of language death, and conceivably increase conversations surrounding language death.

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